



University of
Nottingham
UK | CHINA | MALAYSIA

NEAR AND FAR-FIELD DEFORMATION FROM THE 2023 TURKEY EARTHQUAKES USING GNSS DATA.

Dimitrios Anastasiou¹, Panos Psimoulis², Xanthos Papanikolaou¹, Maria Tsakiri¹

¹National Technical University of Athens, School of Rural, Surveying and Geoinformatics Engineering, Greece (danastasiou@mail.ntua.gr)

²Nottingham Geospatial Institute, University of Nottingham, UK



EUROPEAN GEOSCIENCES UNION GENERAL ASSEMBLY 2025 VIENNA | AUSTRIA | 27 APR. - 02 MAY 2025

INTRODUCTION

On February 6th, 2023 at 01:17:35 UTC the M_w 7.8 Nurdagi-Pazarcik earthquake nucleated ~ 15 km southeast of the mapped trace of the East Anatolian Fault Zone. Relocations place the hypocenter at (37.0234° E, 37.2444° N, depth=12 km) and analyses of teleseismic data show a left-lateral source mechanism on a vertical or near vertical fault. A vigorous aftershock sequence followed and a little over 9 hours after the first event, at 10:24:49 UTC, the M_w 7.6 [1]

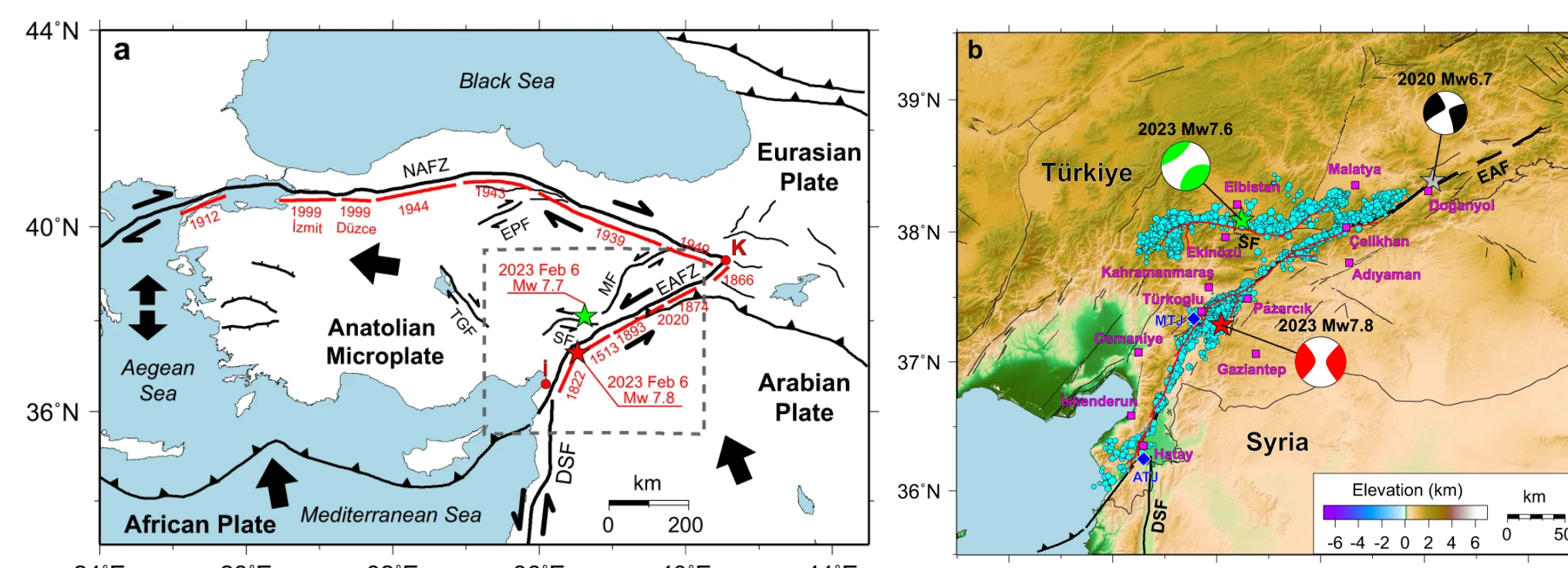
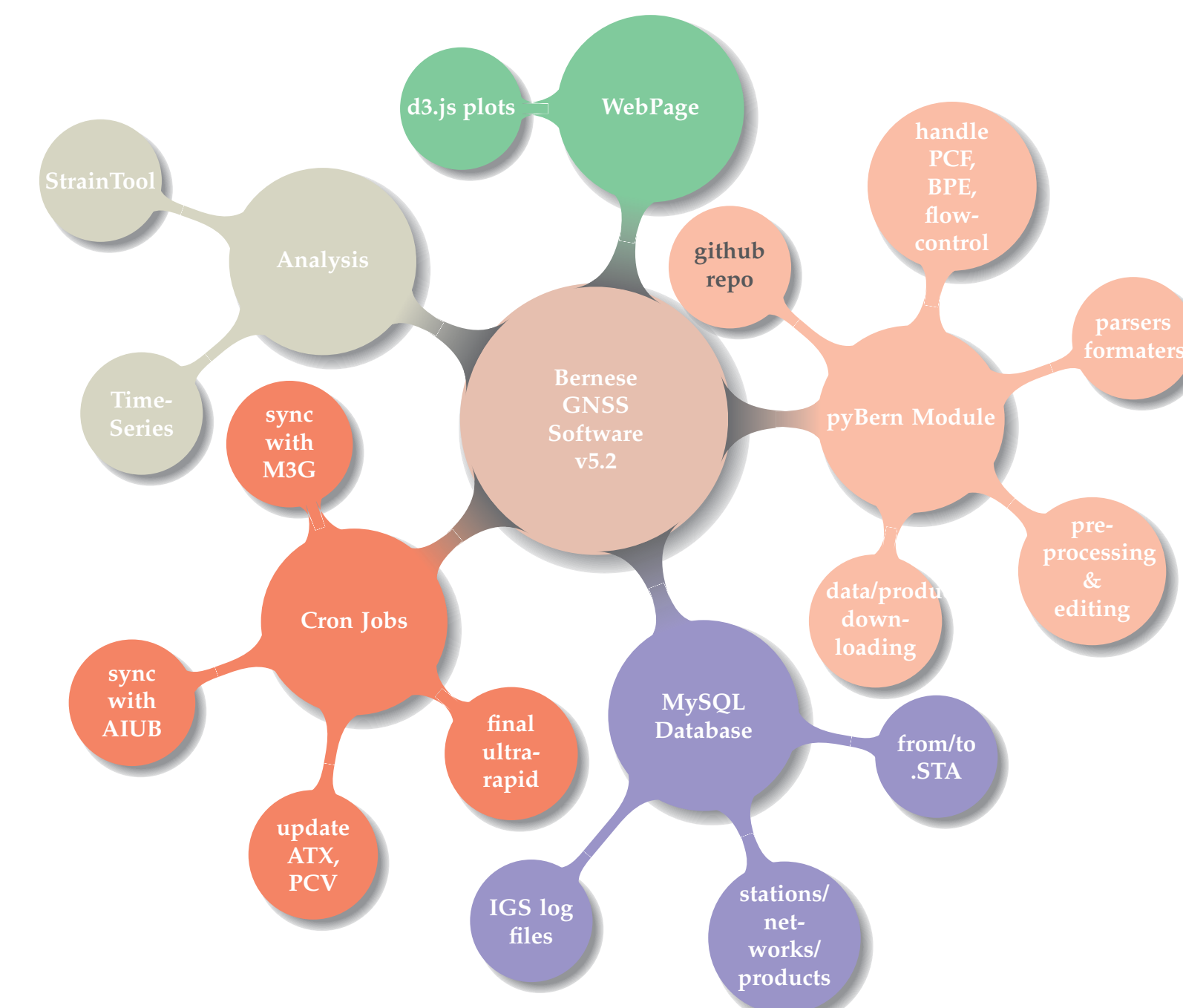


Fig.1: Tectonic background (a) and focal mechanisms (b) of 2023 Turkey earthquake doublet

DATA ANALYSIS

The data described above were analyzed using the Bernese GNSS Software (v. 5.2) (Dach et al., 2015). Both GPS and GLONASS observation were used when available (most of 2021 and 2022). IGB14 was used as the frame of reference, using a set of 19 IGS stations for alignment. CODE final products were incorporated in the analysis, consistent with the frame of choice. The key parameter of interest was station coordinates.



Processing is consistent with EUREF standards (Guidelines for Analysis Centres).

- SINEX with required info/blocks,
- Reference frame **IGb14**,
- IERS Conventions 2010,
- IGS/CODE products,
- ocean loading corrections (**FES2004**),
- 3° elevation cut-off angle; elevation dependent weighting,
- GMF and/or **VMF1**; Chen-Herring gradient parameter,
- ambiguities fixed (length-dependent algorithm),
- use GLONASS obs (when available)
- use ATX files (**epn_14.atx**) - **individual calibrations**

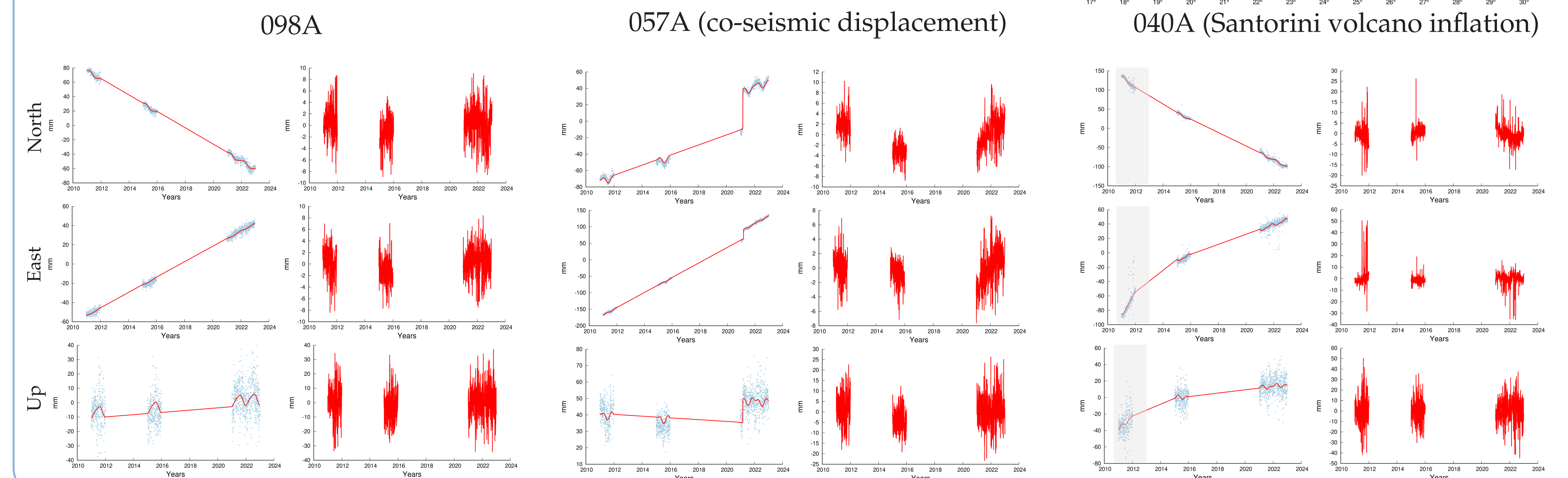
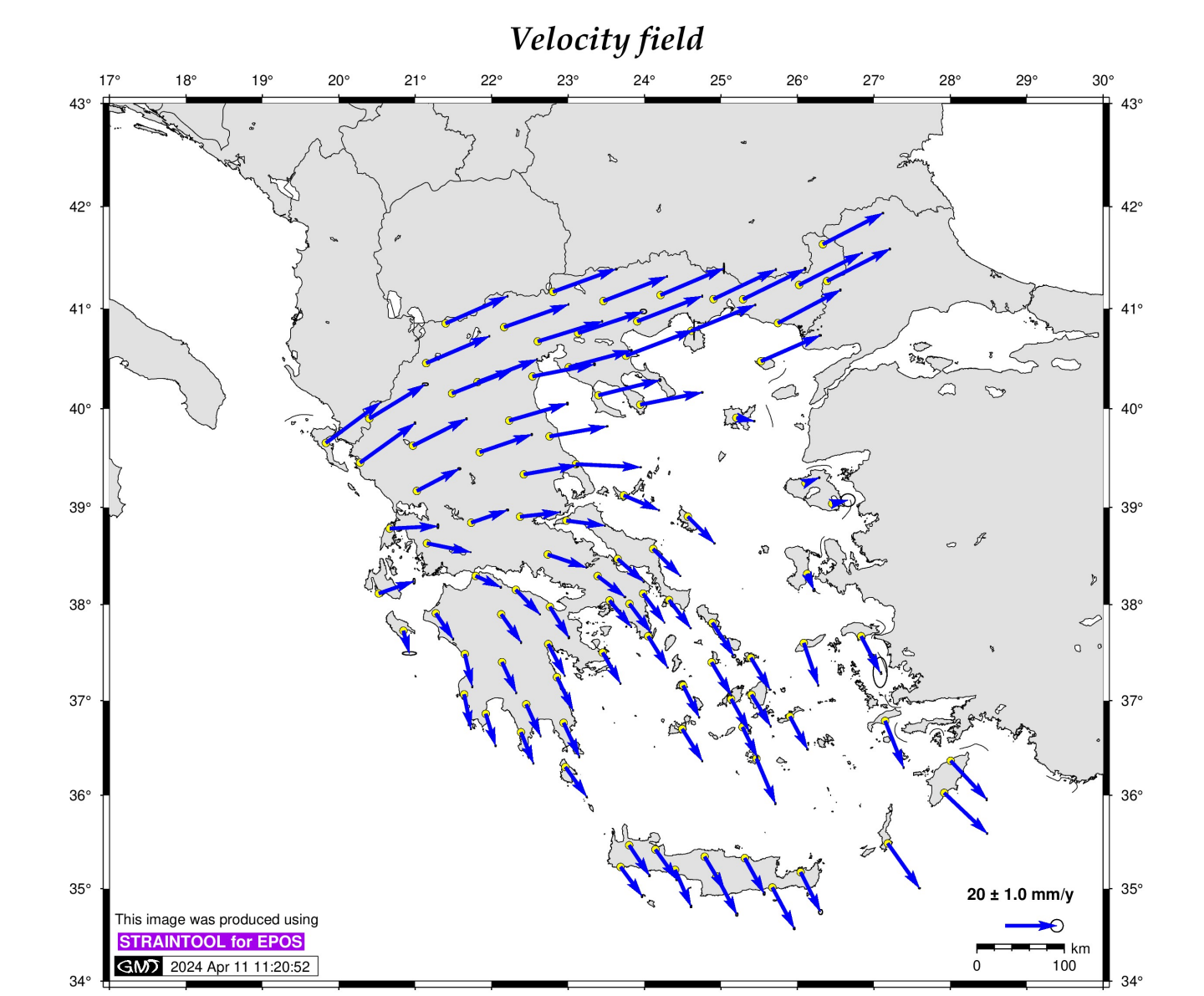
ESTIMATION OF VELOCITY FIELD

Following the estimation of daily site coordinates, position time-series were stacked for every site of the network. The latter were in turn analyzed using the Hector software (Bos et al., 2013) for the estimation of tectonic velocities and harmonic signals. The time span of more than ten years allows estimation of site velocities, yet the temporal density of data could inhibit the accuracy that can be obtained.

Time series analysis of three different station are presented. Velocity field with respect to IGB14 presented on the following map.

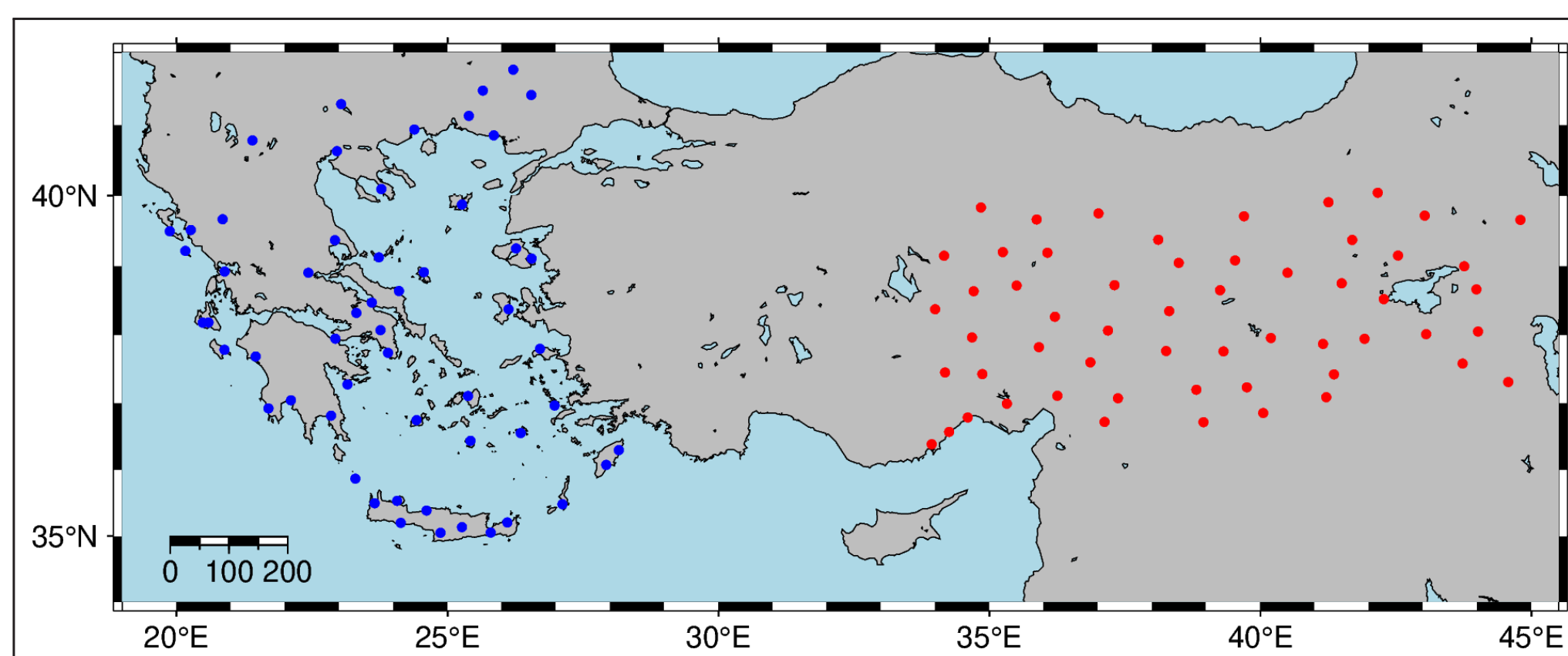
Velocities vary per component:

comp	min (mm/yr)	max (mm/yr)
North	-17.9	15.5
East	2.4	26.0
Up	-5.5	4.7



GNSS STATIONS

Different



REFERENCES

- [1] Diego Melgar, Tuncay Taymaz, Athanassios Ganas, Brendan Crowell, Taylan Öcalan, Metin Kahraman, Varvara Tsironi, Seda Yolsal-Çevikbil, Sotiris Valkaniotis, Tahir Serkan Irmak, Tuna Eken, Ceyhan Erman, Berkan Özkan, Ali Hasan Dogan, and Cemali Altıntaş. Sub- and super-shear ruptures during the 2023 mw 7.8 and mw 7.6 earthquake doublet in se Türkiye. *Seismica*, 2(3), March 2023.
- [2] Chengli Liu, Thorne Lay, Rongjiang Wang, Tuncay Taymaz, Zujun Xie, Xiong Xiong, Tahir Serkan Irmak, Metin Kahraman, and Ceyhan Erman. Complex multi-fault rupture and triggering during the 2023 earthquake doublet in southeastern Türkiye. *Nature Communications*, 14(1), September 2023.
- [3] Jianghui Geng, Xingyu Chen, Yuanxin Pan, Shuyin Mao, Chenghong Li, Jinning Zhou, and Kunlun Zhang. Pride ppp-ar: an open-source software for gps ppp ambiguity resolution. *GPS Solutions*, 23(4), July 2019.



FUTURE RESEARCH

We consider this study to be a work-in-progress; we plan to go on with data analysis for all the available years. Currently, we are working on :

- the analysis of vertical velocities
- the investigation of the deformation field in smaller areas

AKNOWLEDGMENT

We acknowledge support for this research by the project "Support of Permanent GNSS station Network of Hellenic Cadastre" funded by the **Hellenic Cadastre**.

Authors want to thank you Hellenic Cadastre providing the data of HEPOS Network.

